

spark

```
<pyspark.sql.connect.session.SparkSession at 0xff62494682f0>
```

```
from pyspark.sql.types import StructField, StructType, IntegerType,
StringType, BooleanType, DateType, DecimalType
from pyspark.sql.functions import col, when, sum, avg, row_number
from pyspark.sql.window import Window

from pyspark.sql import SparkSession
```

```
#create session
```

```
spark = SparkSession.builder.appName("IPL Data
Analysis").getOrCreate()
```

spark

```
<pyspark.sql.connect.session.SparkSession at 0xff62494682f0>
```

```
ball_by_ball_schema = StructType([
    StructField("match_id", IntegerType(), True),
    StructField("over_id", IntegerType(), True),
    StructField("ball_id", IntegerType(), True),
    StructField("innings_no", IntegerType(), True),
    StructField("team_batting", StringType(), True),
    StructField("team_bowling", StringType(), True),
    StructField("striker_batting_position", IntegerType(), True),
    StructField("extra_type", StringType(), True),
    StructField("runs_scored", IntegerType(), True),
    StructField("extra_runs", IntegerType(), True),
    StructField("wides", IntegerType(), True),
    StructField("legbyes", IntegerType(), True),
    StructField("byes", IntegerType(), True),
    StructField("noballs", IntegerType(), True),
    StructField("penalty", IntegerType(), True),
    StructField("bowler_extras", IntegerType(), True),
    StructField("out_type", StringType(), True),
    StructField("caught", BooleanType(), True),
    StructField("bowled", BooleanType(), True),
    StructField("run_out", BooleanType(), True),
    StructField("lbw", BooleanType(), True),
    StructField("retired_hurt", BooleanType(), True),
    StructField("stumped", BooleanType(), True),
    StructField("caught_and_bowled", BooleanType(), True),
    StructField("hit_wicket", BooleanType(), True),
    StructField("obstructingfeild", BooleanType(), True),
    StructField("bowler_wicket", BooleanType(), True),
    StructField("match_date", DateType(), True),
    StructField("season", IntegerType(), True),
    StructField("striker", IntegerType(), True),
    StructField("non_striker", IntegerType(), True),
```

```

    StructField("bowler", IntegerType(), True),
    StructField("player_out", IntegerType(), True),
    StructField("fielders", IntegerType(), True),
    StructField("striker_match_sk", IntegerType(), True),
    StructField("strikersk", IntegerType(), True),
    StructField("nonstriker_match_sk", IntegerType(), True),
    StructField("nonstriker_sk", IntegerType(), True),
    StructField("fielder_match_sk", IntegerType(), True),
    StructField("fielder_sk", IntegerType(), True),
    StructField("bowler_match_sk", IntegerType(), True),
    StructField("bowler_sk", IntegerType(), True),
    StructField("playerout_match_sk", IntegerType(), True),
    StructField("battingteam_sk", IntegerType(), True),
    StructField("bowlingteam_sk", IntegerType(), True),
    StructField("keeper_catch", BooleanType(), True),
    StructField("player_out_sk", IntegerType(), True),
    StructField("matchdatesk", DateType(), True)
])

ball_by_ball_df =
spark.read.schema(ball_by_ball_schema).format("csv").option("header", "
true").load("s3://ipl-data-analysis-project/Ball_By_Ball.csv")

match_schema = StructType([
    StructField("match_sk", IntegerType(), True),
    StructField("match_id", IntegerType(), True),
    StructField("team1", StringType(), True),
    StructField("team2", StringType(), True),
    StructField("match_date", DateType(), True),
    StructField("season_year", IntegerType(), True),
    StructField("venue_name", StringType(), True),
    StructField("city_name", StringType(), True),
    StructField("country_name", StringType(), True),
    StructField("toss_winner", StringType(), True),
    StructField("match_winner", StringType(), True),
    StructField("toss_name", StringType(), True),
    StructField("win_type", StringType(), True),
    StructField("outcome_type", StringType(), True),
    StructField("manofmach", StringType(), True),
    StructField("win_margin", IntegerType(), True),
    StructField("country_id", IntegerType(), True)
])

match_df =
spark.read.schema(match_schema).format("csv").option("header", "true").
load("s3://ipl-data-analysis-project/Match.csv")

player_schema = StructType([
    StructField("player_sk", IntegerType(), True),
    StructField("player_id", IntegerType(), True),
    StructField("player_name", StringType(), True),

```

```

    StructField("dob", DateType(), True),
    StructField("batting_hand", StringType(), True),
    StructField("bowling_skill", StringType(), True),
    StructField("country_name", StringType(), True)
])

player_df =
spark.read.schema(player_schema).format("csv").option("header","true")
.load("s3://ipl-data-analysis-project/Player.csv")

player_match_schema = StructType([
    StructField("player_match_sk", IntegerType(), True),
    StructField("playermatch_key", DecimalType(), True),
    StructField("match_id", IntegerType(), True),
    StructField("player_id", IntegerType(), True),
    StructField("player_name", StringType(), True),
    StructField("dob", DateType(), True),
    StructField("batting_hand", StringType(), True),
    StructField("bowling_skill", StringType(), True),
    StructField("country_name", StringType(), True),
    StructField("role_desc", StringType(), True),
    StructField("player_team", StringType(), True),
    StructField("opposit_team", StringType(), True),
    StructField("season_year", IntegerType(), True),
    StructField("is_manofthematch", BooleanType(), True),
    StructField("age_as_on_match", IntegerType(), True),
    StructField("isplayers_team_won", BooleanType(), True),
    StructField("batting_status", StringType(), True),
    StructField("bowling_status", StringType(), True),
    StructField("player_captain", StringType(), True),
    StructField("opposit_captain", StringType(), True),
    StructField("player_keeper", StringType(), True),
    StructField("opposit_keeper", StringType(), True)
])

player_match_df =
spark.read.schema(player_match_schema).format("csv").option("header","
true").load("s3://ipl-data-analysis-project/Player_match.csv")

team_schema = StructType([
    StructField("team_sk", IntegerType(), True),
    StructField("team_id", IntegerType(), True),
    StructField("team_name", StringType(), True)
])

team_df =
spark.read.schema(team_schema).format("csv").option("header","true").l
oad("s3://ipl-data-analysis-project/Team.csv")

```

```

# Filter to include only valid deliveries (excluding extras like wides
and no balls for specific analyses)
ball_by_ball_df = ball_by_ball_df.filter((col("wides") == 0) &
(col("noballs")==0))

# Aggregation: Calculate the total and average runs scored in each
match and inning
total_and_avg_runs = ball_by_ball_df.groupBy("match_id",
"innings_no").agg(
    sum("runs_scored").alias("total_runs"),
    avg("runs_scored").alias("average_runs")
)

# Window Function: Calculate running total of runs in each match for
each over
windowSpec =
Window.partitionBy("match_id","innings_no").orderBy("over_id")

ball_by_ball_df = ball_by_ball_df.withColumn(
    "running_total_runs",
    sum("runs_scored").over(windowSpec)
)

# Conditional Column: Flag for high impact balls (either a wicket or
more than 6 runs including extras)
ball_by_ball_df = ball_by_ball_df.withColumn(
    "high_impact",
    when((col("runs_scored") + col("extra_runs") > 6) |
(col("bowler_wicket") == True), True).otherwise(False)
)

ball_by_ball_df.show(5)

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+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
|match_id|over_id|ball_id|innings_no|team_batting|team_bowling|
striker_batting_position|extra_type|runs_scored|extra_runs|wides|
legbyes|byes|noballs|penalty|bowler_extras|out_type|caught|
bowled|run_out|lbw|retired_hurt|stumped|caught_and_bowled|hit_wicket|
obstructingfeild|bowler_wicket|match_date|season|striker|non_striker|
bowler|player_out|fielders|striker_match_sk|strikersk|
nonstriker_match_sk|nonstriker_sk|fielder_match_sk|fielder_sk|

```

```

bowler_match_sk|bowler_sk|playerout_match_sk|battingteam_sk|
bowlingteam_sk|keeper_catch|player_out_sk|matchdatesk|
running_total_runs|high_impact|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
| 335987| 1| 1| 2| 2| 1|
1| No Extras| 1| 0| 0| 0| 0|
0| 0|Not Applicable| NULL| NULL| NULL|NULL|
NULL| NULL| NULL| NULL| NULL| NULL|
NULL| NULL| 2008| 6| 7| 106| NULL| NULL|
12694| 5| 12695| 6| -1|
-1| 12715| 105| -1| 1|
0| NULL| 0| NULL| 3|
false|
| 335987| 1| 3| 2| 2| 1|
2| No Extras| 0| 0| 0| 0| 0|
0| 0|Not Applicable| NULL| NULL| NULL|NULL|
NULL| NULL| NULL| NULL| NULL| NULL|
NULL| NULL| 2008| 7| 6| 106| NULL| NULL|
12695| 6| 12694| 5| -1|
-1| 12715| 105| -1| 1|
0| NULL| 0| NULL| 3|
false|
| 335987| 1| 4| 2| 2| 1|
2| No Extras| 1| 0| 0| 0| 0|
0| 0|Not Applicable| NULL| NULL| NULL|NULL|
NULL| NULL| NULL| NULL| NULL| NULL|
NULL| NULL| 2008| 7| 6| 106| NULL| NULL|
12695| 6| 12694| 5| -1|
-1| 12715| 105| -1| 1|
0| NULL| 0| NULL| 3|
false|
| 335987| 1| 5| 2| 2| 1|
1| No Extras| 1| 0| 0| 0| 0|
0| 0|Not Applicable| NULL| NULL| NULL|NULL|
NULL| NULL| NULL| NULL| NULL| NULL|
NULL| NULL| 2008| 6| 7| 106| NULL| NULL|
12694| 5| 12695| 6| -1|
-1| 12715| 105| -1| 1|
0| NULL| 0| NULL| 3|
false|

```

```

| 335987| 1| 6| 2| 2| 1|
2| No Extras| 0| 0| 0| 0| 0|
0| 0|Not Applicable| NULL| NULL| NULL|NULL|
NULL| NULL| NULL| NULL| NULL|
NULL| NULL| 2008| 7| 6| 106| NULL| NULL|
12695| 6| 12694| 5| -1|
-1| 12715| 105| -1| 1|
0| NULL| 0| NULL| 3|
false|

```

```

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+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
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+-----+-----+-----+-----+-----+-----+

```

only showing top 5 rows

```

from pyspark.sql.functions import year, month, dayofmonth, when

```

```

# Extracting year, month, and day from the match date for more
detailed time-based analysis

```

```

match_df = match_df.withColumn("year", year("match_date"))
match_df = match_df.withColumn("month", month("match_date"))
match_df = match_df.withColumn("day", dayofmonth("match_date"))

```

```

# High margin win: categorizing win margins into 'high', 'medium', and
'low'

```

```

match_df = match_df.withColumn(
    "win_margin_category",
    when(col("win_margin") >= 100, "High")
    .when((col("win_margin") >= 50) & (col("win_margin") < 100),
"Medium")
    .otherwise("Low")
)

```

```

# Analyze the impact of the toss: who wins the toss and the match

```

```

match_df = match_df.withColumn(
    "toss_match_winner",
    when(col("toss_winner") == col("match_winner"),
"Yes").otherwise("No")
)

```

```

# Show the enhanced match DataFrame

```

```

match_df.show(2)

```

```

+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+
|match_sk|match_id|team1|team2|
match_date|season_year|venue_name|city_name|country_name|
toss_winner|match_winner|toss_name|win_type|outcome_type|
manofmach|win_margin|country_id|year|month|day|win_margin_category|
toss_match_winner|
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+
|0|335987|Royal Challengers...|Kolkata Knight Ri...|
NULL|2008|M Chinnaswamy Sta...|Bangalore|India|Royal
Challengers...|Kolkata Knight Ri...|field|runs|Result|BB
McCullum|140|1|NULL|NULL|NULL|High|
No|
|1|335988|Kings XI Punjab|Chennai Super Kings|
NULL|2008|Punjab Cricket As...|Chandigarh|India|Chennai
Super Kings|Chennai Super Kings|bat|runs|Result|MEK
Hussey|33|1|NULL|NULL|NULL|Low|
Yes|
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+
only showing top 2 rows

```

```

from pyspark.sql.functions import lower, regexp_replace

# Normalize and clean player names
player_df = player_df.withColumn("player_name",
lower(regexp_replace("player_name", "[^a-zA-Z0-9 ]", "")))

# Handle missing values in 'batting_hand' and 'bowling_skill' with a
default 'unknown'
player_df = player_df.na.fill({"batting_hand": "unknown",
"bowling_skill": "unknown"})

# Categorizing players based on batting hand
player_df = player_df.withColumn(
    "batting_style",
    when(col("batting_hand").contains("left"), "Left-
Handed").otherwise("Right-Handed")
)

```

```
# Show the modified player DataFrame
```

```
player_df.show(2)
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|player_sk|player_id|player_name| dob|  batting_hand|  bowling_skill|
country_name|batting_style|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
|          0|          1| sc ganguly|NULL| Left-hand bat|Right-arm medium|
India| Right-Handed|
|          1|          2|bb mccullum|NULL|Right-hand bat|Right-arm medium|
New Zealand| Right-Handed|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+
only showing top 2 rows
```

```
from pyspark.sql.functions import col, when, current_date, expr
```

```
# Add a 'veteran_status' column based on player age
```

```
player_match_df = player_match_df.withColumn(
    "veteran_status",
    when(col("age_as_on_match") >= 35, "Veteran").otherwise("Non-
Veteran")
)
```

```
# Dynamic column to calculate years since debut
```

```
player_match_df = player_match_df.withColumn(
    "years_since_debut",
    (year(current_date()) - col("season_year"))
)
```

```
# Show the enriched DataFrame
```

```
player_match_df.show()
```

```
+-----+-----+-----+-----+-----+-----+
+---+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+
|player_match_sk|playermatch_key|match_id|player_id|  player_name|
dob|  batting_hand|  bowling_skill|country_name|role_desc|
player_team|  opposit_team|season_year|is_manofthetmatch|
age_as_on_match|isplayers_team_won|batting_status|bowling_status|
player_captain|opposit_captain|player_keeper|opposit_keeper|
veteran_status|years_since_debut|
+-----+-----+-----+-----+-----+-----+
+---+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
```

```

+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|          -1|          -1|          -1|          -1|          N/A|
NULL|          NULL|          NULL|          NULL|          NULL|
NULL|          NULL|          NULL|          NULL|          NULL|
NULL|          NULL|          NULL|          NULL|          NULL|
NULL|          NULL|          NULL|          Non-Veteran|          NULL|
|          12694|          NULL|          335987|          6|          R Dravid|
NULL|Right-hand bat| Right-arm offbreak|          India| Captain|Royal
Challengers...|Kolkata Knight Ri...|          2008|          NULL|
35|          NULL|          NULL|          NULL|          R Dravid|
SC Ganguly| MV Boucher|          WP Saha|          Veteran|
18|
|          12695|          NULL|          335987|          7|          W Jaffer|
NULL|Right-hand bat| Right-arm offbreak|          India| Player|Royal
Challengers...|Kolkata Knight Ri...|          2008|          NULL|
30|          NULL|          NULL|          NULL|          R Dravid|
SC Ganguly| MV Boucher|          WP Saha|          Non-Veteran|
18|
|          12696|          NULL|          335987|          8|          V Kohli|
NULL|Right-hand bat| Right-arm medium|          India| Player|Royal
Challengers...|Kolkata Knight Ri...|          2008|          NULL|
20|          NULL|          NULL|          NULL|          R Dravid|
SC Ganguly| MV Boucher|          WP Saha|          Non-Veteran|
18|
|          12697|          NULL|          335987|          9|          JH Kallis|
NULL|Right-hand bat|Right-arm fast-me...|South Africa| Player|Royal
Challengers...|Kolkata Knight Ri...|          2008|          NULL|
33|          NULL|          NULL|          NULL|          R Dravid|
SC Ganguly| MV Boucher|          WP Saha|          Non-Veteran|
18|
|          12698|          NULL|          335987|          10|          CL White|
NULL|Right-hand bat| Legbreak googly|          Australia| Player|Royal
Challengers...|Kolkata Knight Ri...|          2008|          NULL|
25|          NULL|          NULL|          NULL|          R Dravid|
SC Ganguly| MV Boucher|          WP Saha|          Non-Veteran|
18|
|          12699|          NULL|          335987|          11|          MV Boucher|
NULL|Right-hand bat| Right-arm medium|South Africa| Keeper|Royal
Challengers...|Kolkata Knight Ri...|          2008|          NULL|
32|          NULL|          NULL|          NULL|          R Dravid|
SC Ganguly| MV Boucher|          WP Saha|          Non-Veteran|
18|
|          12700|          NULL|          335987|          12|          B Akhil|
NULL|Right-hand bat|Right-arm medium-...|          India| Player|Royal
Challengers...|Kolkata Knight Ri...|          2008|          NULL|
31|          NULL|          NULL|          NULL|          R Dravid|
SC Ganguly| MV Boucher|          WP Saha|          Non-Veteran|

```

|                      |                      |                      |             |          |             |
|----------------------|----------------------|----------------------|-------------|----------|-------------|
| 18                   |                      |                      |             |          |             |
|                      | 12701                | NULL                 | 335987      | 13       | AA Noffke   |
| NULL                 | Right-hand bat       | Right-arm fast-me... | Australia   | Player   | Royal       |
| Challengers...       | Kolkata Knight Ri... |                      | 2008        | NULL     |             |
| 31                   | NULL                 | NULL                 | NULL        | R Dravid |             |
| SC Ganguly           | MV Boucher           | WP Saha              | Non-Veteran |          |             |
| 18                   |                      |                      |             |          |             |
|                      | 12702                | NULL                 | 335987      | 14       | P Kumar     |
| NULL                 | Right-hand bat       | Right-arm medium     | India       | Player   | Royal       |
| Challengers...       | Kolkata Knight Ri... |                      | 2008        | NULL     |             |
| 22                   | NULL                 | NULL                 | NULL        | R Dravid |             |
| SC Ganguly           | MV Boucher           | WP Saha              | Non-Veteran |          |             |
| 18                   |                      |                      |             |          |             |
|                      | 12703                | NULL                 | 335987      | 15       | Z Khan      |
| NULL                 | Right-hand bat       | Left-arm fast-medium | India       | Player   | Royal       |
| Challengers...       | Kolkata Knight Ri... |                      | 2008        | NULL     |             |
| 30                   | NULL                 | NULL                 | NULL        | R Dravid |             |
| SC Ganguly           | MV Boucher           | WP Saha              | Non-Veteran |          |             |
| 18                   |                      |                      |             |          |             |
|                      | 12704                | NULL                 | 335987      | 16       | SB Joshi    |
| NULL                 | Left-hand bat        | Slow left-arm ort... | India       | Player   | Royal       |
| Challengers...       | Kolkata Knight Ri... |                      | 2008        | NULL     |             |
| 38                   | NULL                 | NULL                 | NULL        | R Dravid |             |
| SC Ganguly           | MV Boucher           | WP Saha              | Veteran     |          |             |
| 18                   |                      |                      |             |          |             |
|                      | 12705                | NULL                 | 335987      | 1        | SC Ganguly  |
| NULL                 | Left-hand bat        | Right-arm medium     | India       | Captain  |             |
| Kolkata Knight Ri... | Royal Challengers... |                      | 2008        |          |             |
| NULL                 | 36                   | NULL                 | NULL        | NULL     |             |
| SC Ganguly           | R Dravid             | WP Saha              | MV Boucher  |          |             |
| Veteran              | 18                   |                      |             |          |             |
|                      | 12706                | NULL                 | 335987      | 2        | BB McCullum |
| NULL                 | Right-hand bat       | Right-arm medium     | New Zealand | Player   |             |
| Kolkata Knight Ri... | Royal Challengers... |                      | 2008        |          |             |
| NULL                 | 27                   | NULL                 | NULL        | NULL     |             |
| SC Ganguly           | R Dravid             | WP Saha              | MV Boucher  | Non-     |             |
| Veteran              | 18                   |                      |             |          |             |
|                      | 12707                | NULL                 | 335987      | 3        | RT Ponting  |
| NULL                 | Right-hand bat       | Right-arm medium     | Australia   | Player   |             |
| Kolkata Knight Ri... | Royal Challengers... |                      | 2008        |          |             |
| NULL                 | 34                   | NULL                 | NULL        | NULL     |             |
| SC Ganguly           | R Dravid             | WP Saha              | MV Boucher  | Non-     |             |
| Veteran              | 18                   |                      |             |          |             |
|                      | 12708                | NULL                 | 335987      | 4        | DJ Hussey   |
| NULL                 | Right-hand bat       | Right-arm offbreak   | Australia   | Player   |             |
| Kolkata Knight Ri... | Royal Challengers... |                      | 2008        |          |             |
| NULL                 | 31                   | NULL                 | NULL        | NULL     |             |
| SC Ganguly           | R Dravid             | WP Saha              | MV Boucher  | Non-     |             |
| Veteran              | 18                   |                      |             |          |             |

|                      |                      |                    |            |        |                 |
|----------------------|----------------------|--------------------|------------|--------|-----------------|
|                      | 12709                | NULL               | 335987     | 5      | Mohammad Hafeez |
| NULL                 | Right-hand bat       | Right-arm offbreak | Pakistan   | Player |                 |
| Kolkata Knight Ri... | Royal Challengers... | 2008               |            |        |                 |
| NULL                 | 28                   | NULL               | NULL       | NULL   |                 |
| SC Ganguly           | R Dravid             | WP Saha            | MV Boucher | Non-   |                 |
| Veteran              | 18                   |                    |            |        |                 |
|                      | 12710                | NULL               | 335987     | 62     | WP Saha         |
| NULL                 | Right-hand bat       | N/A                | India      | Keeper |                 |
| Kolkata Knight Ri... | Royal Challengers... | 2008               |            |        |                 |
| NULL                 | 24                   | NULL               | NULL       | NULL   |                 |
| SC Ganguly           | R Dravid             | WP Saha            | MV Boucher | Non-   |                 |
| Veteran              | 18                   |                    |            |        |                 |
|                      | 12711                | NULL               | 335987     | 63     | LR Shukla       |
| NULL                 | Right-hand bat       | Right-arm medium   | India      | Player |                 |
| Kolkata Knight Ri... | Royal Challengers... | 2008               |            |        |                 |
| NULL                 | 27                   | NULL               | NULL       | NULL   |                 |
| SC Ganguly           | R Dravid             | WP Saha            | MV Boucher | Non-   |                 |
| Veteran              | 18                   |                    |            |        |                 |
|                      | 12712                | NULL               | 335987     | 82     | AB Agarkar      |
| NULL                 | Right-hand bat       | Right-arm fast     | India      | Player |                 |
| Kolkata Knight Ri... | Royal Challengers... | 2008               |            |        |                 |
| NULL                 | 31                   | NULL               | NULL       | NULL   |                 |
| SC Ganguly           | R Dravid             | WP Saha            | MV Boucher | Non-   |                 |
| Veteran              | 18                   |                    |            |        |                 |

```

+-----+-----+-----+-----+-----+
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```

only showing top 20 rows

```

ball_by_ball_df.createOrReplaceTempView("ball_by_ball")
match_df.createOrReplaceTempView("match")
player_df.createOrReplaceTempView("player")
player_match_df.createOrReplaceTempView("player_match")
team_df.createOrReplaceTempView("team")

```

```
ball_by_ball_df.columns
```

```

['match_id',
 'over_id',
 'ball_id',
 'innings_no',
 'team_batting',
 'team_bowling',
 'striker_batting_position',
 'extra_type',
 'runs_scored',
 'extra_runs',

```

```

'wides',
'legbyes',
'byes',
'noballs',
'penalty',
'bowler_extras',
'out_type',
'caught',
'bowled',
'run_out',
'lbw',
'retired_hurt',
'stumped',
'caught_and_bowled',
'hit_wicket',
'obstructingfeild',
'bowler_wicket',
'match_date',
'season',
'striker',
'non_striker',
'bowler',
'player_out',
'fielders',
'striker_match_sk',
'strikersk',
'nonstriker_match_sk',
'nonstriker_sk',
'fielder_match_sk',
'fielder_sk',
'bowler_match_sk',
'bowler_sk',
'playerout_match_sk',
'battingteam_sk',
'bowlingteam_sk',
'keeper_catch',
'player_out_sk',
'matchdatesk',
'running_total_runs',
'high_impact']

```

```

top_scoring_batsmen_per_season = spark.sql("""
SELECT
p.player_name,
m.season_year,
SUM(b.runs_scored) AS total_runs
FROM ball_by_ball b
JOIN match m ON b.match_id = m.match_id
JOIN player_match pm ON m.match_id = pm.match_id AND b.striker =
pm.player_id

```

```
JOIN player p ON p.player_id = pm.player_id
GROUP BY p.player_name, m.season_year
ORDER BY m.season_year, total_runs DESC
""")
```

```
top_scoring_batsmen_per_season.show(30)
```

| player_name     | season_year | total_runs |
|-----------------|-------------|------------|
| se marsh        | 2008        | 614        |
| g gambhir       | 2008        | 532        |
| st jayasuriya   | 2008        | 508        |
| sr watson       | 2008        | 463        |
| gc smith        | 2008        | 437        |
| ac gilchrist    | 2008        | 431        |
| yk pathan       | 2008        | 430        |
| sk raina        | 2008        | 420        |
| ms dhoni        | 2008        | 414        |
| v sehwag        | 2008        | 399        |
| rg sharma       | 2008        | 399        |
| r dravid        | 2008        | 370        |
| sc ganguly      | 2008        | 349        |
| s dhawan        | 2008        | 340        |
| kc sangakkara   | 2008        | 319        |
| dj hussey       | 2008        | 318        |
| rv uthappa      | 2008        | 316        |
| sa asnodkar     | 2008        | 311        |
| yuvraj singh    | 2008        | 299        |
| pa patel        | 2008        | 297        |
| y venugopal rao | 2008        | 283        |
| ja morkel       | 2008        | 235        |
| mv boucher      | 2008        | 225        |
| jr hopes        | 2008        | 221        |
| am nayar        | 2008        | 205        |
| jh kallis       | 2008        | 199        |
| s badrinath     | 2008        | 192        |
| sp fleming      | 2008        | 192        |
| salman butt     | 2008        | 192        |
| bb mccullum     | 2008        | 188        |

only showing top 30 rows

```
economical_bowlers_powerplay = spark.sql("""
SELECT
p.player_name,
AVG(b.runs_scored) AS avg_runs_per_ball,
COUNT(b.bowler_wicket) AS total_wickets
FROM ball_by_ball b
JOIN player_match pm ON b.match_id = pm.match_id AND b.bowler =
```

```

pm.player_id
JOIN player p ON pm.player_id = p.player_id
WHERE b.over_id <= 6
GROUP BY p.player_name
HAVING COUNT(*) >= 1
ORDER BY avg_runs_per_ball, total_wickets DESC
""")
economical_bowlers_powerplay.show()

```

| player_name     | avg_runs_per_ball  | total_wickets |
|-----------------|--------------------|---------------|
| sm harwood      | 0.3333333333333333 | 0             |
| aj finch        | 0.5                | 0             |
| gr napier       | 0.5                | 0             |
| avesh khan      | 0.5                | 0             |
| a zampa         | 0.5                | 0             |
| ankit soni      | 0.5                | 0             |
| nb singh        | 0.5833333333333334 | 0             |
| ag murtaza      | 0.6538461538461539 | 0             |
| s gopal         | 0.6666666666666666 | 0             |
| sb bangar       | 0.6666666666666666 | 0             |
| d du preez      | 0.6666666666666666 | 0             |
| fh edwards      | 0.6923076923076923 | 0             |
| a kumble        | 0.7685185185185185 | 0             |
| umar gul        | 0.7777777777777778 | 0             |
| kp pietersen    | 0.7777777777777778 | 0             |
| j syed mohammad | 0.7777777777777778 | 0             |
| rj peterson     | 0.8333333333333334 | 0             |
| la carseldine   | 0.8333333333333334 | 0             |
| ss mundhe       | 0.8333333333333334 | 0             |
| tl suman        | 0.8333333333333334 | 0             |

only showing top 20 rows

```

toss_impact_individual_matches = spark.sql("""
SELECT m.match_id, m.toss_winner, m.toss_name, m.match_winner,
       CASE WHEN m.toss_winner = m.match_winner THEN 'Won' ELSE 'Lost'
END AS match_outcome
FROM match m
WHERE m.toss_name IS NOT NULL
ORDER BY m.match_id
""")
toss_impact_individual_matches.show()

```

| match_id | toss_winner | toss_name | match_winner | match_outcome |
|----------|-------------|-----------|--------------|---------------|
|----------|-------------|-----------|--------------|---------------|

```

+-----+
| 335987|Royal Challengers...| field|Kolkata Knight Ri...|
Lost|
| 335988| Chennai Super Kings| bat| Chennai Super Kings|
Won|
| 335989| Rajasthan Royals| bat| Delhi Daredevils|
Lost|
| 335990| Mumbai Indians| bat|Royal Challengers...|
Lost|
| 335991| Deccan Chargers| bat|Kolkata Knight Ri...|
Lost|
| 335992| Kings XI Punjab| bat| Rajasthan Royals|
Lost|
| 335993| Deccan Chargers| bat| Delhi Daredevils|
Lost|
| 335994| Mumbai Indians| field| Chennai Super Kings|
Lost|
| 335995| Rajasthan Royals| field| Rajasthan Royals|
Won|
| 335996| Mumbai Indians| field| Kings XI Punjab|
Lost|
| 335997| Rajasthan Royals| field| Rajasthan Royals|
Won|
| 335998|Kolkata Knight Ri...| bat| Chennai Super Kings|
Lost|
| 335999| Deccan Chargers| field| Deccan Chargers|
Won|
| 336000| Delhi Daredevils| bat| Kings XI Punjab|
Lost|
| 336001| Chennai Super Kings| bat| Chennai Super Kings|
Won|
| 336002|Kolkata Knight Ri...| bat| Mumbai Indians|
Lost|
| 336003|Royal Challengers...| field| Delhi Daredevils|
Lost|
| 336004| Kings XI Punjab| field| Kings XI Punjab|
Won|
| 336005| Rajasthan Royals| bat| Rajasthan Royals|
Won|
| 336006| Chennai Super Kings| bat| Delhi Daredevils|
Lost|

```

```

+-----+-----+-----+-----+-----+-----+

```

```

+-----+

```

only showing top 20 rows

```

average_runs_in_wins = spark.sql("""
SELECT p.player_name, AVG(b.runs_scored) AS avg_runs_in_wins, COUNT(*)
AS innings_played
FROM ball_by_ball b
JOIN player_match pm ON b.match_id = pm.match_id AND b.striker =

```

```
pm.player_id
JOIN player p ON pm.player_id = p.player_id
JOIN match m ON pm.match_id = m.match_id
WHERE m.match_winner = pm.player_team
GROUP BY p.player_name
ORDER BY avg_runs_in_wins ASC
""")
average_runs_in_wins.show()
```

| player_name       | avg_runs_in_wins    | innings_played |
|-------------------|---------------------|----------------|
| jj bumrah         | 0.0                 | 2              |
| j theron          | 0.0                 | 1              |
| i sharma          | 0.0                 | 1              |
| sn thakur         | 0.0                 | 2              |
| a nehra           | 0.0                 | 2              |
| ts mills          | 0.0                 | 3              |
| kp appanna        | 0.0                 | 1              |
| vr aaron          | 0.0                 | 5              |
| anirudh singh     | 0.0                 | 1              |
| t thushara        | 0.2                 | 5              |
| sa abbott         | 0.25                | 4              |
| yashpal singh     | 0.3157894736842105  | 19             |
| kc cariappa       | 0.3333333333333333  | 3              |
| s sreesanth       | 0.3333333333333333  | 3              |
| jd unadkat        | 0.4                 | 5              |
| sm harwood        | 0.42857142857142855 | 7              |
| r shukla          | 0.5                 | 2              |
| sandeep sharma    | 0.5                 | 6              |
| b kumar           | 0.5                 | 10             |
| washington sundar | 0.5                 | 8              |

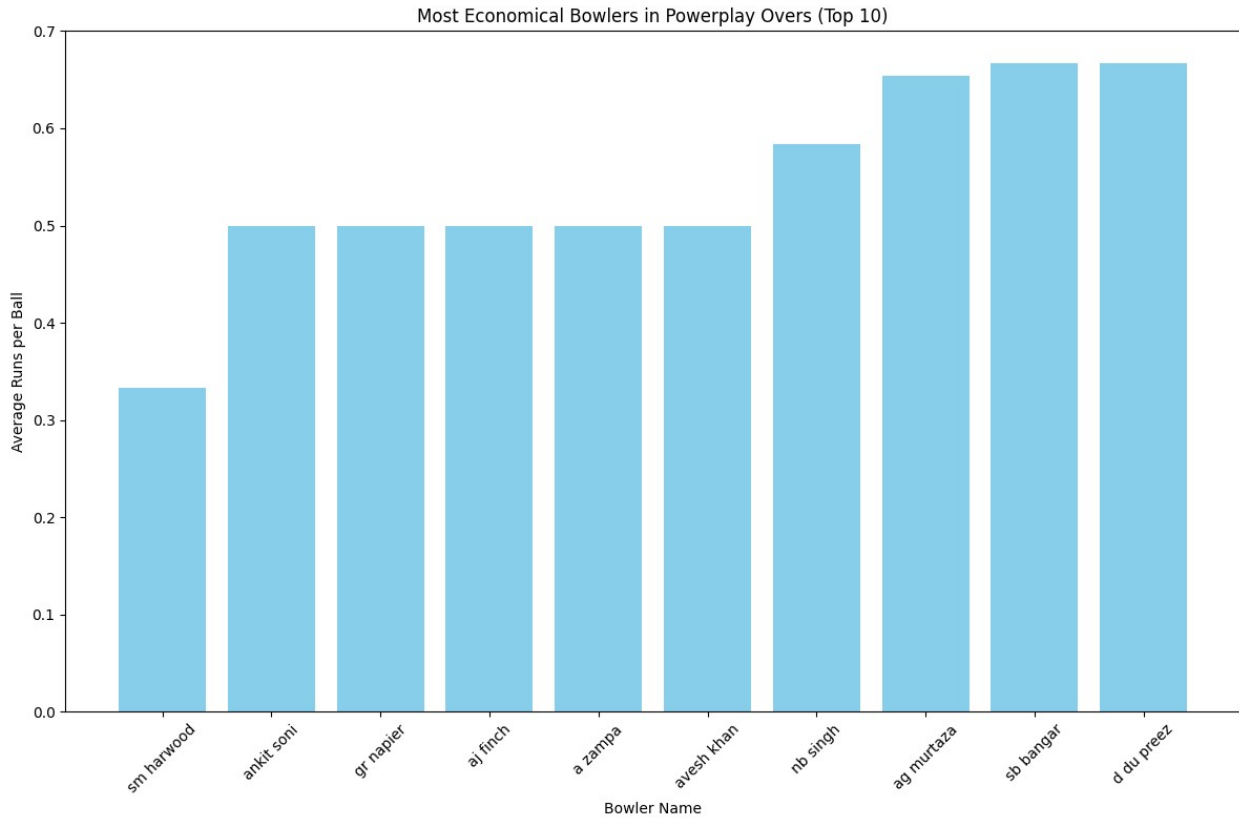
only showing top 20 rows

```
import matplotlib.pyplot as plt

# Assuming 'economical_bowlers_powerplay' is already executed and
# available as a Spark DataFrame
economical_bowlers_pd = economical_bowlers_powerplay.toPandas()

# Visualizing using Matplotlib
plt.figure(figsize=(12, 8))
# Limiting to top 10 for clarity in the plot
top_economical_bowlers = economical_bowlers_pd.nsmallest(10,
'avg_runs_per_ball')
plt.bar(top_economical_bowlers['player_name'],
top_economical_bowlers['avg_runs_per_ball'], color='skyblue')
plt.xlabel('Bowler Name')
plt.ylabel('Average Runs per Ball')
```

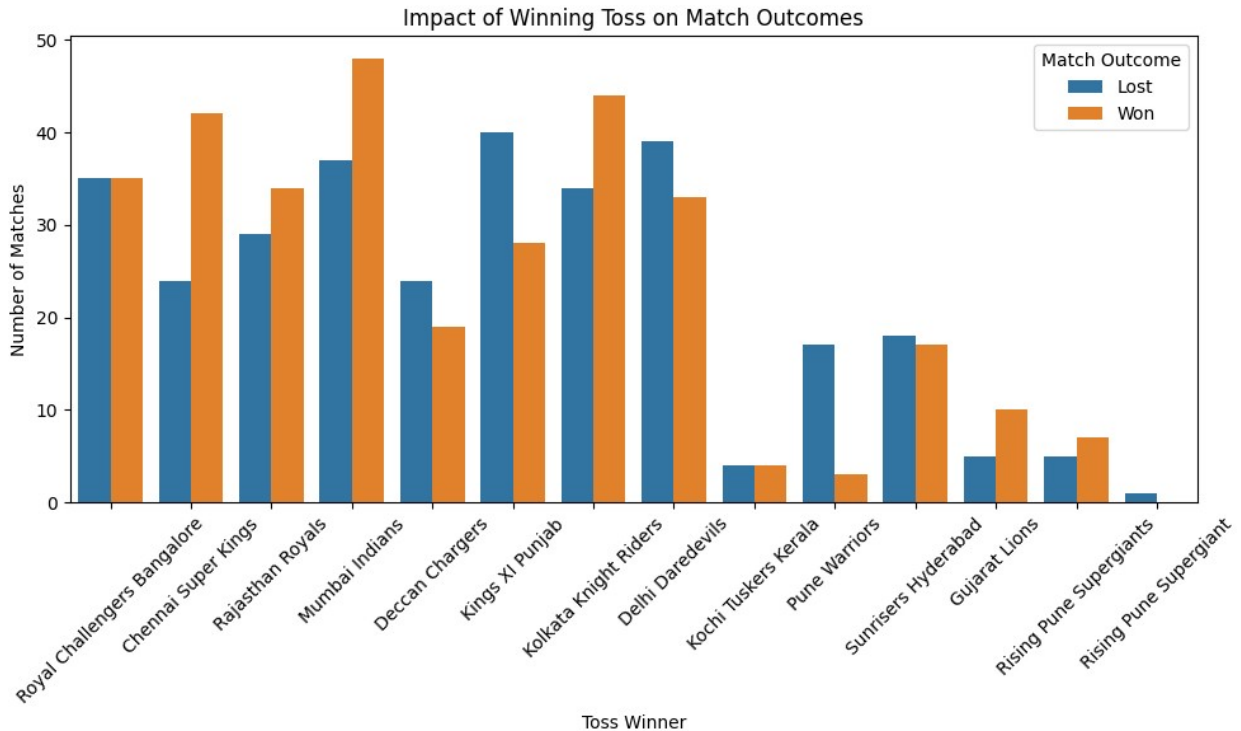
```
plt.title('Most Economical Bowlers in Powerplay Overs (Top 10)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
import seaborn as sns

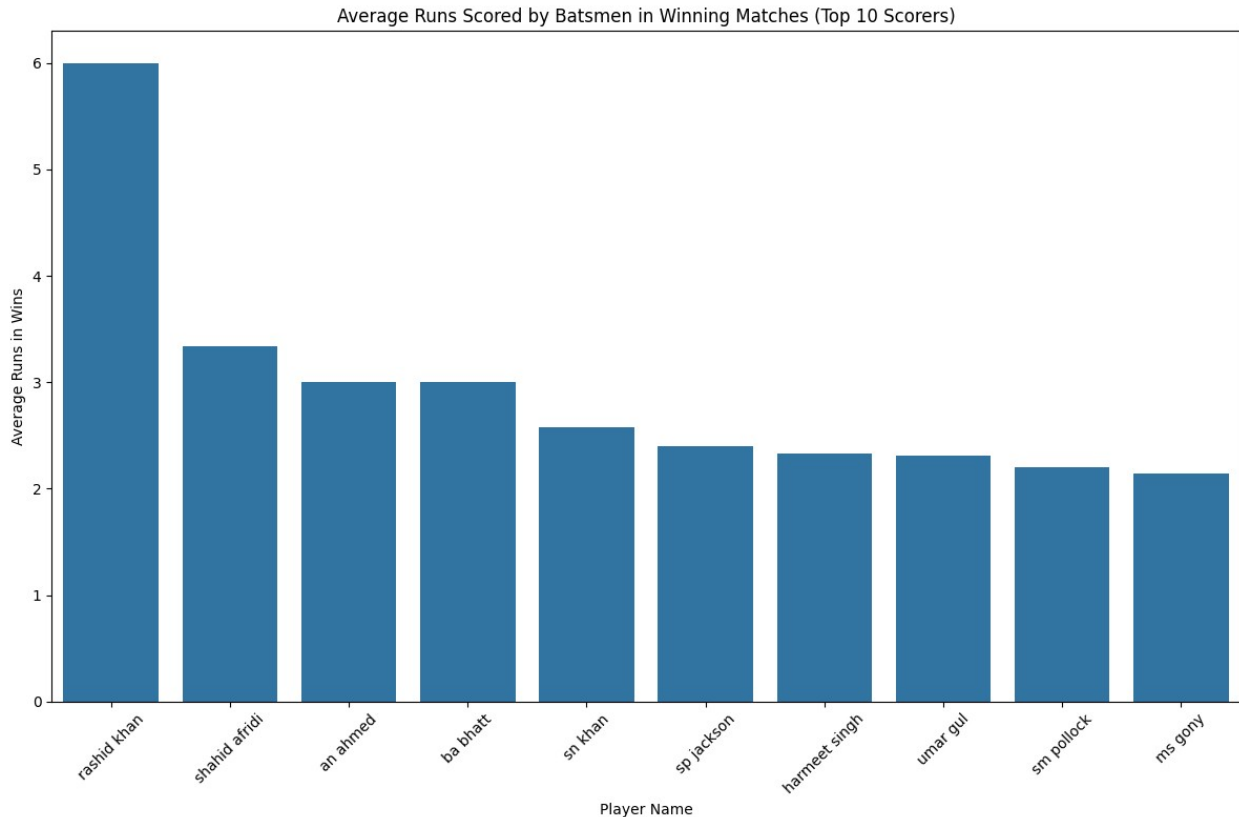
toss_impact_pd = toss_impact_individual_matches.toPandas()

# Creating a countplot to show win/loss after winning toss
plt.figure(figsize=(10, 6))
sns.countplot(x='toss_winner', hue='match_outcome',
data=toss_impact_pd)
plt.title('Impact of Winning Toss on Match Outcomes')
plt.xlabel('Toss Winner')
plt.ylabel('Number of Matches')
plt.legend(title='Match Outcome')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
average_runs_pd = average_runs_in_wins.toPandas()

# Using seaborn to plot average runs in winning matches
plt.figure(figsize=(12, 8))
top_scorers = average_runs_pd.nlargest(10, 'avg_runs_in_wins')
sns.barplot(x='player_name', y='avg_runs_in_wins', data=top_scorers)
plt.title('Average Runs Scored by Batsmen in Winning Matches (Top 10 Scorers)')
plt.xlabel('Player Name')
plt.ylabel('Average Runs in Wins')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

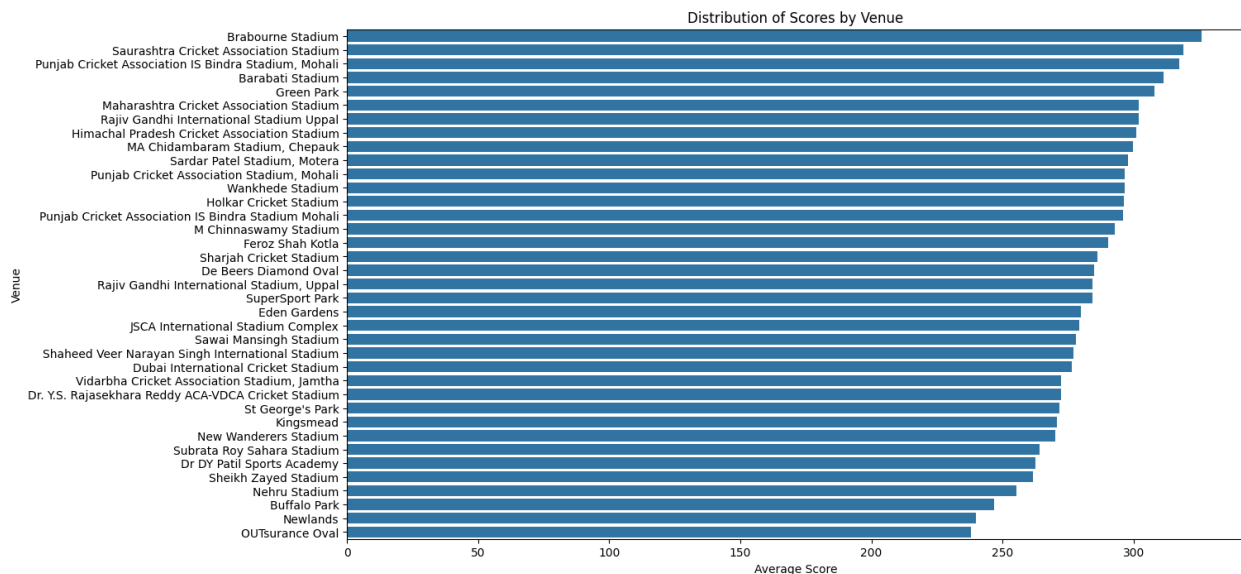


```
# Execute SQL Query
scores_by_venue = spark.sql("""
SELECT venue_name, AVG(total_runs) AS average_score, MAX(total_runs)
AS highest_score
FROM (
    SELECT ball_by_ball.match_id, match.venue_name, SUM(runs_scored)
AS total_runs
    FROM ball_by_ball
    JOIN match ON ball_by_ball.match_id = match.match_id
    GROUP BY ball_by_ball.match_id, match.venue_name
)
GROUP BY venue_name
ORDER BY average_score DESC
""")

# Convert to Pandas DataFrame
scores_by_venue_pd = scores_by_venue.toPandas()

# Plot
plt.figure(figsize=(14, 8))
sns.barplot(x='average_score', y='venue_name',
data=scores_by_venue_pd)
plt.title('Distribution of Scores by Venue')
plt.xlabel('Average Score')
```

```
plt.ylabel('Venue')
plt.show()
```



```
# Execute SQL Query
dismissal_types = spark.sql("""
SELECT out_type, COUNT(*) AS frequency
FROM ball_by_ball
WHERE out_type IS NOT NULL
GROUP BY out_type
ORDER BY frequency DESC
""")
```

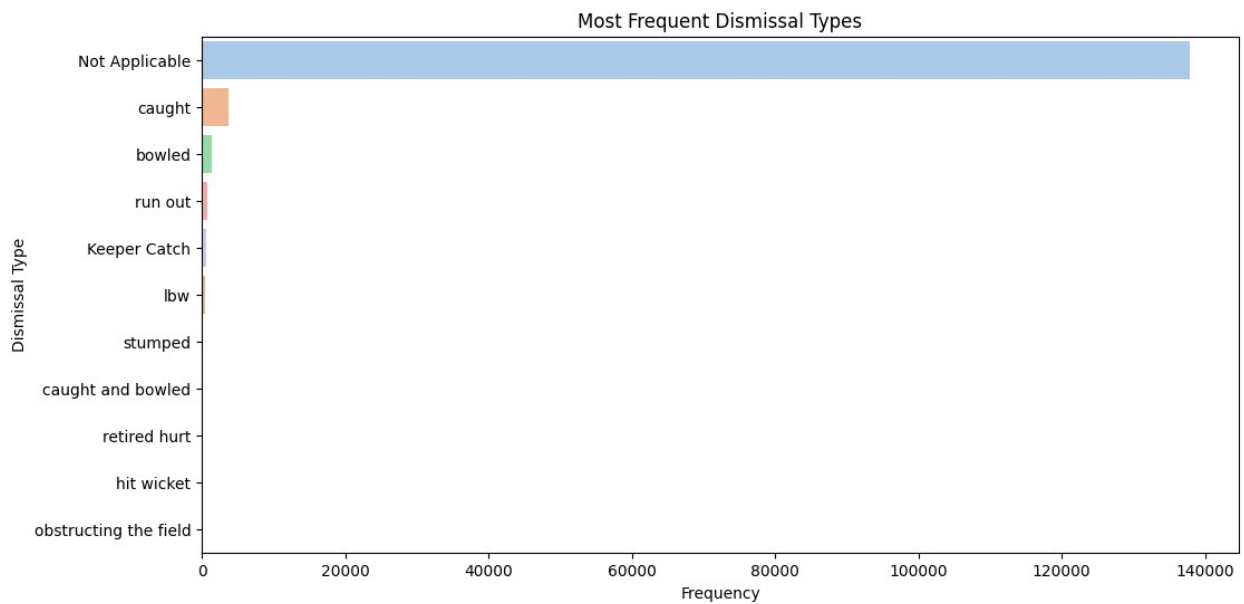
```
# Convert to Pandas DataFrame
dismissal_types_pd = dismissal_types.toPandas()
```

```
# Plot
plt.figure(figsize=(12, 6))
sns.barplot(x='frequency', y='out_type', data=dismissal_types_pd,
palette='pastel')
plt.title('Most Frequent Dismissal Types')
plt.xlabel('Frequency')
plt.ylabel('Dismissal Type')
plt.show()
```

```
/home/spark-86b28bcd-6b32-49e8-b09e-cd/.ipykernel/85/command-
8336789939169600-2176878546:7: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x='frequency', y='out_type', data=dismissal_types_pd,
palette='pastel')
```



```
# Execute SQL Query
team_toss_win_performance = spark.sql("""
SELECT team1, COUNT(*) AS matches_played, SUM(CASE WHEN toss_winner =
match_winner THEN 1 ELSE 0 END) AS wins_after_toss
FROM match
WHERE toss_winner = team1
GROUP BY team1
ORDER BY wins_after_toss DESC
""")

# Convert to Pandas DataFrame
team_toss_win_pd = team_toss_win_performance.toPandas()

# Plot
plt.figure(figsize=(12, 8))
sns.barplot(x='wins_after_toss', y='team1', data=team_toss_win_pd)
plt.title('Team Performance After Winning Toss')
plt.xlabel('Wins After Winning Toss')
plt.ylabel('Team')
plt.show()
```

